Ramsey/Cass/Koopmans Model Cheatsheet

Macroeconomics I - Lecture 2

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1 Notation and Definitions

Core Variables

- K_t: Aggregate capital stock
- L_t: Labor force (population), grows at rate n
- Y_t: Aggregate output
- C_t: Aggregate consumption
- A_t: Technology level (also denoted B_t), grows at rate x
- N_t: Family size at time t, $N_t = N_0 e^{nt}$ (normalize $N_0 = 1$)

Per-Worker Variables

$$k_t = \frac{K_t}{L_t} \qquad \text{(capital per worker)}$$
$$c_t = \frac{C_t}{L_t} \qquad \text{(consumption per worker)}$$
$$y_t = \frac{Y_t}{L_t} \qquad \text{(output per worker)}$$

Per-Effective-Worker Variables (Efficiency Units)

$$\tilde{k}_t = \frac{K_t}{A_t L_t} \qquad \text{(capital per efficiency unit)}$$
$$\tilde{c}_t = \frac{C_t}{A_t L_t} \qquad \text{(consumption per efficiency unit)}$$
$$\tilde{y}_t = \frac{Y_t}{A_t L_t} \qquad \text{(output per efficiency unit)}$$

Parameters

- n: Population growth rate
- x: Technology (TFP) growth rate, $\dot{B}_t/B_t = x$
- δ: Depreciation rate
- ρ: Subjective discount rate ($\rho > 0$)
- σ: CRRA curvature parameter (relative risk aversion)
- α: Capital share in Cobb-Douglas production

Prices

- r_t: Real interest rate (net of depreciation)
- w_t: Real wage rate

2 Model Setup

Environment

- Closed economy (no trade)
- Single homogeneous good (can be consumed or invested)
- Representative household and representative firm
- Three competitive markets: capital, labor, final good
- Infinite-lived households (dynasty)

Production Function

$$Y_t = F(K_t, B_t L_t) \quad \text{or} \quad F(K_t, A_t L_t)$$
$$\text{with CRS:} \quad F(\lambda K, \lambda BL) = \lambda F(K, BL)$$

Technology B_t grows at constant rate:

$$\frac{\dot{B}_t}{B_t} = x \quad \Rightarrow \quad B_t = B_0 e^{xt}$$

For balanced growth, technology must be labor-augmenting (Uzawa 1961).

3 Household Optimization

Household Problem

Representative household chooses $\{c_t, a_t\}$ to maximize:

$$\max_{\{c_t, a_t\}} \left\{ \int_0^\infty u(c_t) e^{-(\rho-n)t} \, \mathrm{d}t \right\}$$

Subject to:

$$c_t + \dot{a}_t + n a_t \leq w_t + r_t a_t$$

with a_0 given and No-Ponzi Game condition:

$$\lim_{t\to\infty} a_t \exp\left(- \int_0^t (r_s - n) \, \mathrm{d}s \right) \geq 0$$

Where $a_t = A_t / N_t$ is assets per worker.

Present-Value Hamiltonian

$$\mathcal{J} = u(c_t) e^{-(\rho-n)t} + \nu_t \left[w_t + (r_t - n) a_t - c_t \right]$$

Necessary FOCs from Hamiltonian:

1. Control variable (c_t):

$$\mathcal{J}_c = 0 \Rightarrow \nu_t = u_c(c_t) e^{-(\rho-n)t} \tag{FOC-C}$$

2. State variable (a_t):

$$\mathcal{J}_a = -\dot{\nu}_t \Rightarrow \dot{\nu}_t = -(r_t - n)\nu_t \tag{FOC-A}$$

3. State evolution:

$$\mathcal{J}_\nu = \dot{a}_t \Rightarrow \dot{a}_t = w_t + (r_t - n) a_t - c_t \tag{State}$$

Euler Equation Derivation

Step 1: Take logs of FOC-C

$$\log \nu_t = \log u_c(c_t) - (\rho - n)t$$

Step 2: Differentiate with respect to t

$$\frac{\dot{\nu}_t}{\nu_t} = \frac{u_{cc}\dot{c}_t}{u_c} - (\rho - n)$$

Step 3: Use FOC-A ($\dot{\nu}_t = -(r_t - n)\nu_t$)

$$-(r_t - n) = \frac{u_{cc}c_t}{u_c}\frac{\dot{c}_t}{c_t} - (\rho - n)$$

Step 4: Rearrange to get Euler Equation

$$\frac{\dot{c}_t}{c_t} = \left[-\frac{u_{cc}(c_t)c_t}{u_c(c_t)}\right]^{-1}(r_t - \rho) \qquad \text{(Euler-General)}$$

The term in brackets is the inverse of the elasticity of intertemporal substitution (EIS).

CRRA Utility Case

For CRRA utility: $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, we have:

$$-\frac{u_{cc}c}{u_c} = \sigma \quad \text{(constant)}$$

Therefore:

$$\boxed{\frac{\dot{c}_t}{c_t} = \frac{1}{\sigma}(r_t - \rho)} \qquad \text{(Euler-CRRA)}$$

Intuition:

- If $r = \rho$: $\dot{c} = 0$, constant consumption
- If $r > \rho$: $\dot{c} > 0$, consumption grows over time
- If $r < \rho$: $\dot{c} < 0$, consumption declines over time
- Higher σ: smaller response of consumption to interest rate differentials

4 Firm Problem and Equilibrium

Firm Optimization

Representative firm rents capital and labor to maximize profits:

$$\Pi_t = F(K_t, B_tL_t) - (r_t + \delta)K_t - w_tL_t$$

Firm FOCs (static problem):

1. Capital:

$$\frac{\partial \Pi}{\partial K} = F_K(K_t, B_t L_t) - (r_t + \delta) = 0 \Rightarrow \boxed{F_K = r_t + \delta} \tag{FOC-K}$$

2. Labor:

$$\frac{\partial \Pi}{\partial L} = F_L(K_t, B_t L_t) - w_t = 0 \Rightarrow \boxed{F_L = w_t} \tag{FOC-L}$$

Note: F_K is MPK (marginal product of capital), F_L is MPL.

Market Clearing

- Labor market: $L_t = N_t$
- Asset market: $K_t = N_t a_t$
- Aggregate resource: $Y_t = C_t + I_t$ where $I_t = \dot{K}_t + \delta K_t$

5 Dynamic System

Capital Accumulation (Per Worker)

From household BC and market clearing:

$$\dot{k}_t = y_t - c_t - (n + \delta) k_t \tag{Capital-Law}$$

This is identical to Solow, except c_t is endogenously determined.

Dynamic System in Efficiency Units

Transform to per-effective-worker terms ($\tilde{k} = k/B$):

$$\dot{\tilde{k}}_t = f(\tilde{k}_t) - \tilde{c}_t - (n + x + \delta)\tilde{k}_t \tag{Dynamics-k}$$

$$\frac{\dot{\tilde{c}}_t}{\tilde{c}_t} + x = \frac{1}{\sigma}\left(f'(\tilde{k}_t) - \delta - \rho\right) \tag{Dynamics-c}$$

Where $f(\tilde{k}) = F(\tilde{k}, 1)$ is the intensive form of the production function.

6 Steady State Analysis

Steady State Conditions

Steady state requires $\dot{\tilde{k}} = 0$ and $\dot{\tilde{c}} = 0$:

$$\dot{\tilde{k}} = 0 \Rightarrow \tilde{c}^* = f(\tilde{k}^*) - (n + x + \delta)\tilde{k}^* \tag{SS-Capital}$$

$$\dot{\tilde{c}} = 0 \Rightarrow f'(\tilde{k}^*) = \delta + \rho + \sigma x \tag{SS-Consumption}$$

Equation (SS-Consumption) is the **Modified Golden Rule**.

Cobb-Douglas Special Case

With $f(\tilde{k}) = \tilde{k}^\alpha$ (Cobb-Douglas):

$$\tilde{k}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{\frac{1}{1-\alpha}} \tag{SS-k-Cobb}$$

$$\tilde{y}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{\frac{\alpha}{1-\alpha}} \tag{SS-y-Cobb}$$

$$\tilde{c}^* = \tilde{y}^* - (n + x + \delta)\tilde{k}^* \tag{SS-c-Cobb}$$

Comparison: Modified Golden Rule vs. Golden Rule

- **Golden Rule (Solow):** $f'(\tilde{k}_{GR}) = n + x + \delta$
- **Modified Golden Rule (Ramsey):** $f'(\tilde{k}^*) = \delta + \rho + \sigma x$

Since $\rho > 0$ and $\sigma > 0$:

$$f'(\tilde{k}^*) = \delta + \rho + \sigma x > \delta + n + x = f'(\tilde{k}_{GR})$$

Because f' is decreasing, we have:

$$\boxed{\tilde{k}^* < \tilde{k}_{GR}} \tag{Key-Insight}$$

The Ramsey steady state always has less capital than the Golden Rule.

7 Transversality Condition and Dynamic Efficiency

TVC Statement

$$\lim_{t\to\infty} \nu_t a_t = 0 \tag{TVC}$$

With CRRA utility and equilibrium conditions:

Transforming the TVC:
Using $\nu_t = \tilde{c}_t^{-\sigma} e^{-(\rho-n)t}$ and $a_t = \tilde{k}_t B_t$:

$$\lim_{t\to\infty} \tilde{c}_t^{-\sigma} \tilde{k}_t B_t e^{-(\rho-n)t} = 0$$

Since $\tilde{k}, \tilde{c}$ are bounded in steady state and $B_t = B_0 e^{xt}$:

$$\lim_{t\to\infty} e^{(n+(1-\sigma)x-\rho)t} = 0$$

Therefore:

$$\boxed{n + (1-\sigma)x - \rho < 0} \qquad \text{(TVC-Condition)}$$

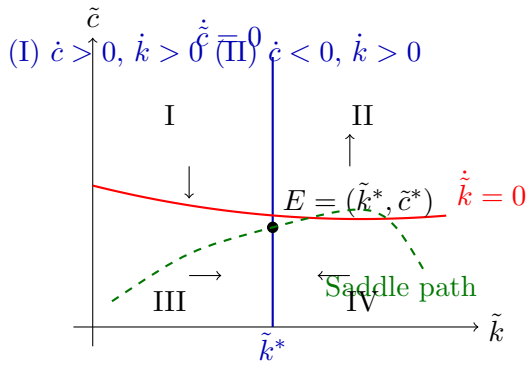
Implications of TVC

1. **Dynamic Efficiency:** The Ramsey equilibrium is always dynamically efficient
2. **Interest Rate:** In steady state, $r^* = \rho + \sigma x$
3. **Capital Level:** $\tilde{k}^* < \tilde{k}_{GR}$ (cannot over-accumulate)
4. **Savings Rate:** $s^* = \alpha \frac{n+x+\delta}{\rho+\sigma x+\delta} < \alpha$

The TVC rules out Ponzi schemes and ensures the household does not leave positive assets at infinity.

8 Phase Diagram and Transitional Dynamics

Phase Diagram: Key Loci



Region	$\dot{\tilde{c}}$	$\dot{\tilde{k}}$
I (above both loci)	< 0	> 0
II (left of $\dot{c} = 0$, below $\dot{k} = 0$)	> 0	> 0
III (below both loci)	> 0	< 0
IV (right of $\dot{c} = 0$, above $\dot{k} = 0$)	< 0	< 0

Stability Analysis: Saddle Path

Saddle-Path Stability:
For any initial $\tilde{k}_0$, there is a unique $\tilde{c}_0$ on the stable arm that leads to the steady state.

- **Above saddle path**: $\tilde{c} > \tilde{c}_{sp} \Rightarrow \dot{\tilde{c}} < 0$ and $\dot{\tilde{k}} < 0 \Rightarrow$ eventually $k \to 0$ (violates feasibility)
- **Below saddle path**: $\tilde{c} < \tilde{c}_{sp} \Rightarrow \dot{\tilde{c}} > 0$ and $\dot{\tilde{k}} > 0 \Rightarrow$ eventually violates TVC ($k \to \infty$)

Only the saddle path satisfies both the Euler equation and the TVC.

9 Comparative Statics

Parameter Changes and Steady State Effects

Parameter	$\tilde{k}^*$	$\tilde{c}^*$	$\tilde{y}^*$	r^*	Savings Rate
$\rho \uparrow$ (more impatient)	$\downarrow$	$\downarrow$	$\downarrow$	—	$\downarrow$
$\sigma \uparrow$ (more risk averse)	$\downarrow$	$\downarrow$	$\downarrow$	—	$\downarrow$
$\delta \uparrow$	$\uparrow$	$\downarrow$	ambiguous	—	$\uparrow$
$n \uparrow$	no effect	no effect	no effect	—	no effect
$x \uparrow$ (tech growth)	$\downarrow$	$\downarrow$	no effect	$\uparrow$	$\downarrow$

Intuition for $\rho \uparrow$ (or $\sigma \uparrow$):

1. The $\dot{\tilde{c}} = 0$ locus shifts left (lower $\tilde{k}$ for given $\tilde{c}$)
2. New steady state: lower $\tilde{k}^*$, lower $\tilde{c}^*$, lower $\tilde{y}^*$
3. Households are more impatient, save less, accumulate less capital
4. Cannot distinguish ρ and σ from steady-state data alone

Key Insight on Population Growth:
In Ramsey (unlike Solow), n does NOT affect the steady-state levels of $\tilde{k}^*$, $\tilde{c}^*$, $\tilde{y}^*$.
The determination of $\tilde{k}^*$ comes directly from the Euler equation, where n plays no role.

10 Problem-Solving Templates

Template 1: Household Optimization (Hamiltonian Method)

1. **Set up objective:** $\max \int_0^\infty u(c_t)e^{-(\rho-n)t}\mathrm{d}t$
2. **Write BC:** $c_t + \dot{a}_t + na_t = w_t + r_t a_t$
3. **Form Hamiltonian:** $\mathcal{J} = u(c)e^{-(\rho-n)t} + \nu[w + (r-n)a - c]$

4. **FOCs:**
   - $u_c(c)e^{-(\rho-n)t} = \nu$
   - $\dot{\nu} = -(r-n)\nu$
5. **Eliminate ν:** Take logs, differentiate, combine to get Euler equation
6. **Add TVC:** $\lim_{t\to\infty} \nu_t a_t = 0$

Template 2: Find Steady State

1. Write dynamic system in per-efficient-worker form:

$$\dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n+x+\delta)\tilde{k}$$
$$\dot{\tilde{c}} = \frac{\tilde{c}}{\sigma}[f'(\tilde{k}) - \delta - \rho - \sigma x]$$

2. Set $\dot{\tilde{k}} = 0$ and $\dot{\tilde{c}} = 0$
3. Solve for $\tilde{k}^*$: $f'(\tilde{k}^*) = \delta + \rho + \sigma x$
4. Solve for $\tilde{c}^*$: $\tilde{c}^* = f(\tilde{k}^*) - (n+x+\delta)\tilde{k}^*$
5. For Cobb-Douglas: $\tilde{k}^* = \left[\frac{\alpha}{\delta+\rho+\sigma x}\right]^{1/(1-\alpha)}$

Template 3: Phase Diagram Analysis

1. Draw $\dot{\tilde{c}} = 0$ locus: vertical line at $\tilde{k}^*$ where $f'(\tilde{k}^*) = \delta + \rho + \sigma x$
2. Draw $\dot{\tilde{k}} = 0$ locus: $\tilde{c} = f(\tilde{k}) - (n+x+\delta)\tilde{k}$ (inverted U)
3. Find intersection: steady state $E = (\tilde{k}^*, \tilde{c}^*)$
4. Identify stable arm (saddle path): negatively sloped curve through E
5. Add arrows: use sign charts for each region
6. Analyze dynamics: convergence only along saddle path

Template 4: Policy Shock Analysis (Comparative Statics)

1. Identify which loci shift:
   - ρ, σ, x affect $\dot{\tilde{c}} = 0$ locus
   - n, x, δ affect $\dot{\tilde{k}} = 0$ locus
2. Determine direction of shift
3. Find new steady state (intersection of shifted loci)
4. Discuss transition: does economy converge? What happens to variables?

11 Common Pitfalls

1. **Confusing ρ with r:** ρ is subjective discount rate (patience), r is market interest rate. They are equal only in steady state.

2. **Forgetting technology growth x:** In the accumulation equation, use $(n + x + \delta)$, not just $(n + \delta)$.

3. **Mixing per-worker vs. per-effective variables:** Always transform: $k = \tilde{k} \cdot B$. Growth rates differ by x.

4. **Forgetting TVC:** Many students omit the transversality condition, but it's essential for dynamic efficiency and ruling out Ponzi schemes.

5. **Phase diagram errors:**

   - $\dot{c} = 0$ locus is VERTICAL (depends only on k)
   - $\dot{k} = 0$ locus is INVERTED U (depends on both k and c)

6. **Signs in Euler equation:** Remember $\dot{c}/c = (1/\sigma)(r - \rho)$, not $(r + \rho)$.

7. **Golden Rule confusion:**

   - Solow GR: $f'(k_{GR}) = n + \delta$
   - Ramsey GR: $f'(k^*) = \delta + \rho + \sigma x$

8. **Discount factor in Hamiltonian:** Remember to use $e^{-(\rho-n)t}$, not $e^{-\rho t}$, when population growth exists.

12 Quick Reference Formulas

$$
\begin{aligned}
\text{Euler Equation:} \quad & \frac{\dot{c}}{c} = \frac{1}{\sigma}(r - \rho) \\
\text{Capital Law of Motion:} \quad & \dot{k} = y - c - (n + \delta)k \\
\text{Per-Efficient Capital:} \quad & \dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n + x + \delta)\tilde{k} \\
\text{Modified Golden Rule:} \quad & f'(\tilde{k}^*) = \delta + \rho + \sigma x \\
\text{SS Interest Rate:} \quad & r^* = \rho + \sigma x \\
\text{Cobb-Douglas SS:} \quad & \tilde{k}^* = \left[\frac{\alpha}{\delta + \rho + \sigma x}\right]^{1/(1-\alpha)} \\
\text{TVC Condition:} \quad & n + (1 - \sigma)x - \rho < 0 \\
\text{SS Savings Rate:} \quad & s^* = \alpha \frac{n + x + \delta}{\rho + \sigma x + \delta}
\end{aligned}
$$

13 Key Formulas Summary Table

Concept	Formula	Equation
Household BC	$c + \dot{a} + na = w + ra$	(1)
Euler (CRRA)	$\dot{c}/c = (r - \rho)/\sigma$	(2)
MPK condition	$F_K = r + \delta$	(3)
Capital law	$\dot{k} = y - c - (n + \delta)k$	(4)
Per-efficient	$\dot{\tilde{k}} = f(\tilde{k}) - \tilde{c} - (n + x + \delta)\tilde{k}$	(5)
Modified GR	$f'(\tilde{k}^*) = \delta + \rho + \sigma x$	(6)
SS interest	$r^* = \rho + \sigma x$	(7)
TVC	$n + (1 - \sigma)x < \rho$	(8)

14 Ramsey vs. Solow Comparison

Feature	Solow Model	Ramsey Model
Consumption decision	Exogenously given ($c = (1 - s)y$)	Endogenous (Euler equation)
Savings rate	Fixed s	Determined by optimization
Steady state k^*	$s/(n + \delta)$ weighted avg	From Euler: $f'(k^*) = \delta + \rho + \sigma x$
Role of n in k^*	Critical	No effect (in efficiency units)
Dynamic efficiency	May be inefficient (too much k)	Always efficient (TVC)
Golden Rule	$f'(k_{GR}) = n + \delta$	$f'(k^*) = \delta + \rho + \sigma x$
Transition	Monotonic	Saddle path dynamics

15 Implementation Notes

Compilation

Compile with: `pdflatex -interaction=nonstopmode Ramsey_CheatSheet.tex`

Packages Used

- amsmath, amssymb, mathtools: Mathematics typesetting
- graphicx: Graphics support
- xcolor: Color definitions
- siunitx: Number formatting
- cleveref: Cross-references
- hyperref: Hyperlinks
- tcolorbox: Colored derivation and result boxes
- booktabs: Professional tables
- enumitem: List formatting